DO NOW — RECALL

Lesson 059 · Thinking Abstractly: What's the Model?

Name: _____ Class: _____ Date: _____

1. What makes a good field in a record?

2. Why give a unique record key?

3. What is a precondition?

4. Trace binary search for 4 in the sorted, zero-indexed array [2,4,6,8,10]. Start with low=0 and high=4. Use mid=(low+high) DIV 2, high=mid-1 or low=mid+1. State the indexes checked, in order.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Information needed for the model/use case.

2. Model answer

To identify instances unambiguously.

3. Model answer

Assumption that must hold for a calculation to be valid.

4. Model answer

2, 0, 1. Values checked are 6, 2, 4. The search finds 4 at index 1.

By the end, you can…

• Abstraction selects only the information required for the task and deliberately excludes irrelevant real-world detail.

• An abstract model is a simplification of reality: here distance and child/adult status determine the fare; hair colour does not.

• The model is only as good as its assumptions. If peak pricing or concessions apply, the abstraction needs refinement.

Vocabulary — know these terms

abstraction — Focus on relevant details.

model — Simplified representation of reality.

assumption — Condition accepted within a model.

irrelevant — Detail that does not affect task.

NEW LEARNING

Read, annotate and explain

Understand

Abstraction selects only the information required for the task and deliberately excludes irrelevant real-world detail.

Model

An abstract model is a simplification of reality: here distance and child/adult status determine the fare; hair colour does not.

Evaluate

The model is only as good as its assumptions. If peak pricing or concessions apply, the abstraction needs refinement.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Choose relevant inputs for calculating parking fee at £2 per hour, rounded to whole hours.

Explain

ParkingFee(hours)=hours×£2 for the simplified model; define non-negative hours and rounding rules.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

For an automated bus fare model, identify three relevant and two irrelevant details. [5]

Question 2

Define inputs, processing and output for library overdue-fee model at 20p/day capped at £2. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate the risks of abstracting a school attendance intervention down to attendance percentage alone. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An abstraction must include every observable detail of reality.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.